LE Growth AI — Complete Replit Build Prompt

> Copy and paste this entire prompt into Replit Agent to generate the full application.

MASTER PROMPT FOR REPLIT AGENT

```
You are a senior full-stack engineer. Build a complete, production-ready,
multi-tenant SaaS application called "LE Growth AI" — an AI-powered social
media management platform for small and medium businesses. Follow every
instruction below exactly.

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## PRODUCT OVERVIEW

Name: LE Growth AI
Tagline: AI-Powered Social Growth
Target customers: Hotels, restaurants, coffee shops, takeaways, hair salons,
guest houses, estate agents, retail shops
Business model: Multi-tenant SaaS — multiple companies use one platform,
fully isolated by company_id

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## TECH STACK

Frontend:
- React 18 + TypeScript
- Vite (build tool)
- TailwindCSS (styling)
- Shadcn UI (component library)
- Lucide React (icons)
- Recharts (charts and analytics)
- React Router DOM v6 (routing)
- React Query / TanStack Query (data fetching)
- date-fns (date handling)

Backend:
- Express.js + TypeScript
- Node.js 20
- OpenAI SDK (AI content generation)
- Supabase JS client (server-side)
- CORS, dotenv, express-rate-limit
```

```
Database + Auth + Storage:
- Supabase (PostgreSQL)
- Supabase Auth (email/password)
- Supabase Storage (media uploads)
- Row Level Security on all tables

Deployment target: Vercel (frontend) + Railway or Vercel serverless (backend)
Source control: GitHub

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## FOLDER STRUCTURE

le-growth-ai/
├── client/                          # React frontend
│   ├── public/
│   │   └── logo.png
│   ├── src/
│   │   ├── assets/
│   │   ├── components/
│   │   │   ├── ui/                  # Shadcn UI components
│   │   │   ├── layout/
│   │   │   │   ├── Sidebar.tsx
│   │   │   │   ├── TopBar.tsx
│   │   │   │   └── AppLayout.tsx
│   │   │   ├── dashboard/
│   │   │   │   ├── StatCard.tsx
│   │   │   │   ├── SocialAccountCard.tsx
│   │   │   │   ├── RecentPostsTable.tsx
│   │   │   │   ├── EngagementChart.tsx
│   │   │   │   └── PlatformGrowthChart.tsx
│   │   │   ├── ai/
│   │   │   │   ├── AIAssistantPanel.tsx
│   │   │   │   └── ContentGeneratorForm.tsx
│   │   │   ├── posts/
│   │   │   │   ├── PostCard.tsx
│   │   │   │   ├── CreatePostForm.tsx
│   │   │   │   └── SchedulePostModal.tsx
│   │   │   └── common/
│   │   │       ├── LoadingSpinner.tsx
│   │   │       ├── EmptyState.tsx
│   │   │       └── ConfirmDialog.tsx
│   │   ├── pages/
│   │   │   ├── auth/
│   │   │   │   ├── LoginPage.tsx
│   │   │   │   ├── SignupPage.tsx
│   │   │   │   └── ForgotPasswordPage.tsx
```

```
│   │   │   ├── DashboardPage.tsx
│   │   │   ├── SocialAccountsPage.tsx
│   │   │   ├── ContentStudioPage.tsx
│   │   │   ├── AIGeneratorPage.tsx
│   │   │   ├── SchedulerPage.tsx
│   │   │   ├── CalendarPage.tsx
│   │   │   ├── AnalyticsPage.tsx
│   │   │   ├── MediaLibraryPage.tsx
│   │   │   ├── SettingsPage.tsx
│   │   │   └── BillingPage.tsx
│   │   ├── hooks/
│   │   │   ├── useAuth.ts
│   │   │   ├── usePosts.ts
│   │   │   ├── useAnalytics.ts
│   │   │   └── useCompany.ts
│   │   ├── lib/
│   │   │   ├── supabase.ts
│   │   │   ├── queryClient.ts
│   │   │   └── utils.ts
│   │   ├── context/
│   │   │   └── AuthContext.tsx
│   │   ├── router/
│   │   │   ├── AppRouter.tsx
│   │   │   └── ProtectedRoute.tsx
│   │   ├── types/
│   │   │   └── index.ts
│   │   ├── App.tsx
│   │   └── main.tsx
│   ├── index.html
│   ├── tailwind.config.ts
│   ├── tsconfig.json
│   └── vite.config.ts
│
├── server/                            # Express backend
│   ├── src/
│   │   ├── routes/
│   │   │   ├── ai.ts
│   │   │   ├── posts.ts
│   │   │   ├── analytics.ts
│   │   │   ├── media.ts
│   │   │   └── companies.ts
│   │   ├── middleware/
│   │   │   ├── auth.ts
│   │   │   └── rateLimit.ts
│   │   ├── services/
│   │   │   ├── openai.service.ts
│   │   │   └── supabase.service.ts
```

````
│   │   └── index.ts
│   ├── package.json
│   └── tsconfig.json
│
├── supabase/
│   ├── schema.sql                    # Full database schema
│   ├── rls.sql                       # Row Level Security policies
│   └── seed.sql                      # Demo seed data
│
├── .env.example
├── .gitignore
├── README.md
└── package.json

---

## DATABASE SCHEMA (supabase/schema.sql)

Create this exact SQL:

```sql
-- Enable UUID extension
CREATE EXTENSION IF NOT EXISTS "uuid-ossp";

-- COMPANIES
CREATE TABLE companies (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  company_name TEXT NOT NULL,
  logo_url TEXT,
  industry TEXT,
  location TEXT,
  website TEXT,
  plan TEXT DEFAULT 'starter' CHECK (plan IN ('starter', 'growth', 'pro',
'enterprise')),
  created_at TIMESTAMPTZ DEFAULT NOW()
);

-- PROFILES (extends Supabase auth.users)
CREATE TABLE profiles (
  id UUID PRIMARY KEY REFERENCES auth.users(id) ON DELETE CASCADE,
  company_id UUID REFERENCES companies(id) ON DELETE CASCADE,
  full_name TEXT,
  email TEXT,
  role TEXT DEFAULT 'user' CHECK (role IN ('admin', 'user')),
  avatar_url TEXT,
  created_at TIMESTAMPTZ DEFAULT NOW()
);
````
````

```sql
-- SOCIAL ACCOUNTS
CREATE TABLE social_accounts (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  company_id UUID NOT NULL REFERENCES companies(id) ON DELETE CASCADE,
  platform TEXT NOT NULL CHECK (platform IN ('instagram', 'facebook',
'tiktok', 'twitter', 'linkedin')),
  username TEXT,
  access_token TEXT,
  refresh_token TEXT,
  followers INTEGER DEFAULT 0,
  posts_count INTEGER DEFAULT 0,
  engagement_rate DECIMAL(5,2) DEFAULT 0,
  connected BOOLEAN DEFAULT FALSE,
  connected_at TIMESTAMPTZ,
  created_at TIMESTAMPTZ DEFAULT NOW(),
  UNIQUE(company_id, platform)
);

-- POSTS
CREATE TABLE posts (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  company_id UUID NOT NULL REFERENCES companies(id) ON DELETE CASCADE,
  platform TEXT NOT NULL CHECK (platform IN ('instagram', 'facebook',
'tiktok', 'twitter', 'linkedin')),
  caption TEXT,
  hashtags TEXT[],
  image_url TEXT,
  status TEXT DEFAULT 'draft' CHECK (status IN ('draft', 'scheduled',
'published', 'failed')),
  scheduled_at TIMESTAMPTZ,
  published_at TIMESTAMPTZ,
  likes INTEGER DEFAULT 0,
  comments INTEGER DEFAULT 0,
  shares INTEGER DEFAULT 0,
  reach INTEGER DEFAULT 0,
  created_by UUID REFERENCES profiles(id),
  created_at TIMESTAMPTZ DEFAULT NOW()
);

-- ANALYTICS (daily snapshots)
CREATE TABLE analytics (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  company_id UUID NOT NULL REFERENCES companies(id) ON DELETE CASCADE,
  platform TEXT NOT NULL,
  date DATE NOT NULL DEFAULT CURRENT_DATE,
  followers INTEGER DEFAULT 0,
```

```
  new_followers INTEGER DEFAULT 0,
  likes INTEGER DEFAULT 0,
  comments INTEGER DEFAULT 0,
  shares INTEGER DEFAULT 0,
  reach INTEGER DEFAULT 0,
  impressions INTEGER DEFAULT 0,
  engagement_rate DECIMAL(5,2) DEFAULT 0,
  created_at TIMESTAMPTZ DEFAULT NOW(),
  UNIQUE(company_id, platform, date)
);

-- MEDIA LIBRARY
CREATE TABLE media (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  company_id UUID NOT NULL REFERENCES companies(id) ON DELETE CASCADE,
  file_name TEXT NOT NULL,
  file_url TEXT NOT NULL,
  file_type TEXT,
  file_size INTEGER,
  uploaded_by UUID REFERENCES profiles(id),
  created_at TIMESTAMPTZ DEFAULT NOW()
);

-- AI GENERATED CONTENT LOG
CREATE TABLE ai_content_log (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  company_id UUID NOT NULL REFERENCES companies(id) ON DELETE CASCADE,
  business_name TEXT,
  industry TEXT,
  location TEXT,
  prompt_type TEXT,
  generated_content JSONB,
  created_by UUID REFERENCES profiles(id),
  created_at TIMESTAMPTZ DEFAULT NOW()
);

-- Indexes for performance
CREATE INDEX idx_posts_company_id ON posts(company_id);
CREATE INDEX idx_posts_status ON posts(status);
CREATE INDEX idx_posts_scheduled_at ON posts(scheduled_at);
CREATE INDEX idx_analytics_company_date ON analytics(company_id, date);
CREATE INDEX idx_media_company_id ON media(company_id);
```

ROW LEVEL SECURITY (supabase/rls.sql)

```sql
-- Enable RLS on all tables
ALTER TABLE companies ENABLE ROW LEVEL SECURITY;
ALTER TABLE profiles ENABLE ROW LEVEL SECURITY;
ALTER TABLE social_accounts ENABLE ROW LEVEL SECURITY;
ALTER TABLE posts ENABLE ROW LEVEL SECURITY;
ALTER TABLE analytics ENABLE ROW LEVEL SECURITY;
ALTER TABLE media ENABLE ROW LEVEL SECURITY;
ALTER TABLE ai_content_log ENABLE ROW LEVEL SECURITY;

-- Helper function: get current user's company_id
CREATE OR REPLACE FUNCTION get_user_company_id()
RETURNS UUID AS $$
  SELECT company_id FROM profiles WHERE id = auth.uid()
$$ LANGUAGE SQL SECURITY DEFINER;

-- Helper function: check if current user is admin
CREATE OR REPLACE FUNCTION is_admin()
RETURNS BOOLEAN AS $$
  SELECT role = 'admin' FROM profiles WHERE id = auth.uid()
$$ LANGUAGE SQL SECURITY DEFINER;

-- COMPANIES policies
CREATE POLICY "Users can view their own company"
  ON companies FOR SELECT
  USING (id = get_user_company_id());

CREATE POLICY "Admins can update their company"
  ON companies FOR UPDATE
  USING (id = get_user_company_id() AND is_admin());

-- PROFILES policies
CREATE POLICY "Users can view profiles in their company"
  ON profiles FOR SELECT
  USING (company_id = get_user_company_id());

CREATE POLICY "Users can update their own profile"
  ON profiles FOR UPDATE
  USING (id = auth.uid());

CREATE POLICY "Enable insert for authenticated users"
  ON profiles FOR INSERT
  WITH CHECK (id = auth.uid());

-- SOCIAL ACCOUNTS policies
CREATE POLICY "Users can view company social accounts"
```

```sql
  ON social_accounts FOR SELECT
  USING (company_id = get_user_company_id());

CREATE POLICY "Admins can manage social accounts"
  ON social_accounts FOR ALL
  USING (company_id = get_user_company_id() AND is_admin());

-- POSTS policies
CREATE POLICY "Users can view company posts"
  ON posts FOR SELECT
  USING (company_id = get_user_company_id());

CREATE POLICY "Users can create posts for their company"
  ON posts FOR INSERT
  WITH CHECK (company_id = get_user_company_id());

CREATE POLICY "Users can update company posts"
  ON posts FOR UPDATE
  USING (company_id = get_user_company_id());

CREATE POLICY "Admins can delete posts"
  ON posts FOR DELETE
  USING (company_id = get_user_company_id() AND is_admin());

-- ANALYTICS policies
CREATE POLICY "Users can view company analytics"
  ON analytics FOR SELECT
  USING (company_id = get_user_company_id());

CREATE POLICY "Admins can insert analytics"
  ON analytics FOR INSERT
  WITH CHECK (company_id = get_user_company_id());

-- MEDIA policies
CREATE POLICY "Users can view company media"
  ON media FOR SELECT
  USING (company_id = get_user_company_id());

CREATE POLICY "Users can upload media for their company"
  ON media FOR INSERT
  WITH CHECK (company_id = get_user_company_id());

CREATE POLICY "Admins can delete media"
  ON media FOR DELETE
  USING (company_id = get_user_company_id() AND is_admin());

-- AI CONTENT LOG policies
```

```sql
CREATE POLICY "Users can view company AI logs"
  ON ai_content_log FOR SELECT
  USING (company_id = get_user_company_id());

CREATE POLICY "Users can create AI logs for their company"
  ON ai_content_log FOR INSERT
  WITH CHECK (company_id = get_user_company_id());

-- Create Supabase Storage bucket
INSERT INTO storage.buckets (id, name, public)
VALUES ('media', 'media', true)
ON CONFLICT DO NOTHING;

CREATE POLICY "Users can upload to company folder"
  ON storage.objects FOR INSERT
  WITH CHECK (bucket_id = 'media' AND auth.uid() IS NOT NULL);

CREATE POLICY "Media is publicly viewable"
  ON storage.objects FOR SELECT
  USING (bucket_id = 'media');
```

AUTO-TRIGGER: Create profile on signup

```sql
CREATE OR REPLACE FUNCTION handle_new_user()
RETURNS TRIGGER AS $$
BEGIN
  INSERT INTO profiles (id, email, full_name)
  VALUES (
    NEW.id,
    NEW.email,
    COALESCE(NEW.raw_user_meta_data->>'full_name', '')
  );
  RETURN NEW;
END;
$$ LANGUAGE plpgsql SECURITY DEFINER;

CREATE TRIGGER on_auth_user_created
  AFTER INSERT ON auth.users
  FOR EACH ROW EXECUTE FUNCTION handle_new_user();
```

```
## ENVIRONMENT VARIABLES (.env.example)
```

Client (prefix with VITE_)

VITE_SUPABASE_URL=your_supabase_project_url
VITE_SUPABASE_ANON_KEY=your_supabase_anon_key
VITE_API_BASE_URL=http://localhost:3001

Server

SUPABASE_URL=your_supabase_project_url
SUPABASE_SERVICE_ROLE_KEY=your_supabase_service_role_key
OPENAI_API_KEY=your_openai_api_key PORT=3001 NODE_ENV=development
ALLOWED_ORIGINS=http://localhost:5173

````
---

## AUTHENTICATION FLOW

### client/src/lib/supabase.ts
```typescript
import { createClient } from '@supabase/supabase-js'

const supabaseUrl = import.meta.env.VITE_SUPABASE_URL
const supabaseAnonKey = import.meta.env.VITE_SUPABASE_ANON_KEY

export const supabase = createClient(supabaseUrl, supabaseAnonKey)
```

### client/src/context/AuthContext.tsx
Create a full AuthContext that:
- Tracks session state using supabase.auth.onAuthStateChange
- Exposes: user, profile, company, loading, signIn, signUp, signOut,
resetPassword
- On login, fetches profile and company from Supabase
- Stores session persistently (Supabase handles this)

### client/src/router/ProtectedRoute.tsx
- Redirects to /login if no session
- Shows loading spinner while checking auth
- Renders children if authenticated

### Auth Pages
LoginPage.tsx:
- Email + password form
- "Remember me" checkbox
- Link to /signup and /forgot-password
- Show LE Growth AI logo at top
- On success, redirect to /dashboard

SignupPage.tsx:
- Full name, email, password, company name fields
- On submit: create Supabase user, then create company record, then update
profile with company_id and role='admin'
- Redirect to /dashboard after signup

ForgotPasswordPage.tsx:
- Email field
- Calls supabase.auth.resetPasswordForEmail()
- Shows success message
````

SIDEBAR NAVIGATION

Create a collapsible left sidebar with these items:

Section: MAIN
- 🏠 Dashboard → /dashboard
- 📊 Analytics → /analytics
- 📈 Reports → /reports

Section: CONTENT
- ✏️ Create Post → /content-studio
- ✨ AI Generator → /ai-generator
- 📅 Content Calendar → /calendar
- ⏰ Scheduled Posts → /scheduler
- 📂 Media Library → /media

Section: SOCIAL ACCOUNTS
- 📷 Instagram → /social-accounts
- 📘 Facebook → /social-accounts
- 🎵 TikTok → /social-accounts

Section: ACCOUNT
- ⚙️ Settings → /settings
- 💳 Billing → /billing

Bottom: User profile avatar + name + company name + logout button

Sidebar should show "LE Growth AI" logo at top. Use the brand colors: primary blue #2563EB, dark sidebar #0F172A, white text.

DASHBOARD PAGE (DashboardPage.tsx)

Replicate the layout from the reference screenshot exactly:

TOP STAT CARDS (4 cards in a row):
1. Total Followers — sum across all platforms — icon: Users — blue accent
2. Total Engagement — sum of likes + comments — icon: Heart — green accent
3. Total Posts — count from posts table — icon: FileText — yellow accent
4. Avg Engagement Rate — average across platforms — icon: BarChart2 — purple accent

Each stat card shows:

```
- Icon in colored circle
- Large number (formatted with commas)
- % change vs last week (green up arrow / red down arrow)
- Label "vs last week"

SOCIAL ACCOUNTS SECTION (3 cards in a row):
For each platform (Instagram, Facebook, TikTok):
- Platform icon + name
- Green "Connected" badge OR grey "Not Connected" badge
- Followers count
- Posts count
- Engagement rate %
- 3 recent media thumbnails (grey placeholder if no media)
- "View [Platform]" button with arrow

AI ASSISTANT PANEL (right sidebar, fixed):
- Header: "AI Assistant" + BETA badge in purple
- Subtext: "Generate content, hashtags, and grow your audience"
- Button: "Generate Post Ideas" (purple, full width)
- Button: "Suggest Hashtags" (outlined, full width)
- Chat input: "Ask me anything..." with send icon
- Quick action chips: "Content ideas", "Best posting times", "Hashtag
suggestions"

RECENT POSTS SECTION:
- Grid of 4 post cards
- Each shows: platform icon, image thumbnail, caption preview, status badge
(Published/Scheduled/Draft), date, engagement metrics (likes, comments,
shares)
- "View All Posts" link

ENGAGEMENT OVERVIEW CHART (bottom right):
- Line chart using Recharts
- 3 lines: Instagram (pink), Facebook (blue), TikTok (dark)
- X-axis: days of week
- Y-axis: engagement count
- Legend below
- "This Week" dropdown

---

## AI CONTENT GENERATOR PAGE (AIGeneratorPage.tsx)

FORM (left panel):
Business name input
Industry dropdown:
  - Hotel / Guest House
```

```
  - Coffee Shop
  - Restaurant
  - Takeaway / Fast Food
  - Hair Salon / Barber
  - Retail Shop
  - Estate Agent / Property
  - Gym / Fitness
  - Spa / Beauty
  - Other

Location input (city/area)
Tone selector: Professional / Friendly / Excited / Informative / Promotional
Post goal: Brand Awareness / Drive Sales / Get Bookings / Increase Followers

PRESET TEMPLATE BUTTONS:
Show 7 quick-fill buttons:
🏨 Hotel  ☕ Coffee Shop  🍕 Restaurant  🥡 Takeaway  💇 Hair Salon  🛍️
Retail  🏠 Estate Agent

Clicking a template pre-fills the form with example data for that business
type.

"Generate Content" button (blue, full width, with sparkle icon)

OUTPUT PANEL (right panel):
Shows 4 tab sections after generation:
1. Instagram Caption
2. Facebook Post
3. TikTok Caption
4. Hashtags + Call to Action

Each section shows:
- Generated text in a card
- "Copy" button
- "Use in Post" button (opens Content Studio with this content pre-filled)
- Character count

MONTHLY CONTENT CALENDAR GENERATOR:
Below the main generator:
- "Generate 30-Day Content Plan" button
- Generates a structured content calendar with post ideas for each day
- Displays as a readable list grouped by week

---

## EXPRESS API SERVER (server/src/index.ts + routes)
```

````
### server/src/index.ts
```typescript
import express from 'express'
import cors from 'cors'
import dotenv from 'dotenv'
import rateLimit from 'express-rate-limit'
import aiRoutes from './routes/ai'
import postsRoutes from './routes/posts'
import analyticsRoutes from './routes/analytics'
import mediaRoutes from './routes/media'

dotenv.config()

const app = express()
const PORT = process.env.PORT || 3001

app.use(cors({ origin: process.env.ALLOWED_ORIGINS?.split(',') }))
app.use(express.json({ limit: '10mb' }))

const limiter = rateLimit({ windowMs: 15 * 60 * 1000, max: 100 })
app.use('/api/', limiter)

app.use('/api/ai', aiRoutes)
app.use('/api/posts', postsRoutes)
app.use('/api/analytics', analyticsRoutes)
app.use('/api/media', mediaRoutes)

app.get('/health', (req, res) => res.json({ status: 'ok', service: 'LE Growth
AI API' }))

app.listen(PORT, () => console.log(`LE Growth AI server running on port
${PORT}`))
```

### server/src/routes/ai.ts
POST /api/ai/generate-content
- Body: { businessName, industry, location, tone, goal }
- Validate all fields present
- Call OpenAI GPT-4o
- Return: { instagram, facebook, tiktok, hashtags, callToAction }

POST /api/ai/generate-hashtags
- Body: { industry, platform, location }
- Return: { hashtags: string[] } — 20-30 relevant hashtags

POST /api/ai/generate-calendar
- Body: { businessName, industry, month }
````

```
- Return: { calendar: Array<{ day, platform, postIdea, caption, hashtags }> }

POST /api/ai/improve-caption
- Body: { caption, platform, tone }
- Return: { improved: string }

### OpenAI Prompts (server/src/services/openai.service.ts)

For generate-content, use this system prompt:
```

You are an expert social media manager for small businesses. Generate engaging, platform-specific content that drives real results. Always match the brand voice, use relevant emojis appropriately, and write captions that encourage action.

Return ONLY valid JSON in this exact format: { "instagram": "caption text with emojis and line breaks", "facebook": "longer more conversational post", "tiktok": "short punchy caption with trending hooks", "hashtags": ["hashtag1", "hashtag2", ...30 hashtags], "callToAction": "specific CTA for this business" }

````
For generate-calendar, return:
```json
{
  "calendar": [
    {
      "week": 1,
      "day": "Monday",
      "date": "2024-06-03",
      "platform": "instagram",
      "theme": "Brand Story",
      "postIdea": "Share your origin story",
      "caption": "Full caption here...",
      "hashtags": ["tag1", "tag2"]
    }
  ]
}
```

---

## CONTENT STUDIO PAGE (ContentStudioPage.tsx)

Split layout:

LEFT: Post Creator
- Platform selector tabs: Instagram | Facebook | TikTok (can select multiple)
- Image upload area (drag and drop + click) — uploads to Supabase Storage
- Caption textarea with character counter (30/2200 for Instagram etc.)
- Hashtags input (shows as tags)
- AI caption suggestions button
- Status selector: Draft / Schedule
- If Schedule: date/time picker
- "Save Draft" button
- "Schedule Post" button
- "Publish Now" button (placeholder — shows "Connect account to publish")

RIGHT: Preview Panel
- Live preview of how post will look on selected platform
- Platform-specific frame (Instagram square, Facebook wider, TikTok portrait)
- Shows image + caption + hashtags as they type

---

## ANALYTICS PAGE (AnalyticsPage.tsx)
````

```
DATE RANGE PICKER at top (last 7 days / 30 days / 90 days / custom)

SUMMARY CARDS (6 cards):
- Total Followers
- New Followers This Period
- Total Likes
- Total Comments
- Total Reach
- Avg Engagement Rate

CHARTS SECTION:

1. Follower Growth (Line chart)
   - 3 lines: Instagram / Facebook / TikTok
   - X-axis: dates
   - Y-axis: follower count

2. Engagement by Platform (Bar chart)
   - Grouped bars per week
   - Shows likes + comments + shares per platform

3. Post Performance (Horizontal bar chart)
   - Top 5 posts by engagement
   - Shows post caption preview + engagement score

4. Best Posting Times (Heatmap grid)
   - Days of week vs hours of day
   - Color intensity = engagement level

PLATFORM BREAKDOWN TABLE:
| Platform | Followers | Posts | Avg Likes | Avg Comments | Engagement Rate |
Each row has platform icon and colored badge.

---

## CALENDAR PAGE (CalendarPage.tsx)

Toggle: Week View / Month View

Month View:
- Full calendar grid
- Each day cell shows scheduled/published posts as colored dots (color =
platform)
- Click day to see posts for that day
- Click post to edit
- "+" button on each day to create post for that date
- Platform legend at top
```

```
Week View:
- 7-column grid with time slots
- Posts appear as colored blocks in their time slot
- Drag to reschedule (if possible, otherwise button-based)

MINI CALENDAR sidebar for navigation
Upcoming posts list sidebar showing next 5 scheduled posts

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## SOCIAL ACCOUNTS PAGE (SocialAccountsPage.tsx)

3 large cards for Instagram, Facebook, TikTok

Each card shows:
- Platform logo + name
- Status: Connected (green) or Not Connected (grey)
- If connected: username, followers, posts, last synced
- If not connected: "Connect [Platform]" button
- If connected: "Disconnect" button + "Refresh Data" button

Below the cards:
- Info box explaining Phase 2 integration coming soon
- "Join Waitlist" button for early access

Manual follower input:
- If not connected via API, allow manual entry of follower counts
- These feed into analytics charts

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## MEDIA LIBRARY PAGE (MediaLibraryPage.tsx)

TOP: Upload zone (drag-and-drop, accepts jpg/png/gif/mp4)
Uploads to Supabase Storage under: media/{company_id}/{filename}

GRID VIEW of all uploaded media:
- 4 columns on desktop, 2 on mobile
- Each item: thumbnail + filename + file size + upload date
- Hover: "Use in Post" button + "Delete" button

FILTER BAR: All / Images / Videos
SEARCH: by filename

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```

```
## SETTINGS PAGE (SettingsPage.tsx)

Tabs: Company Profile | My Profile | Password | Notifications | Team

Company Profile tab:
- Company name input
- Industry dropdown
- Location input
- Website input
- Logo upload (stores in Supabase Storage)
- Save button

My Profile tab:
- Full name input
- Email (read-only)
- Avatar upload
- Role (read-only)
- Save button

Password tab:
- Current password
- New password
- Confirm new password
- Uses supabase.auth.updateUser()

Team tab (Admin only):
- List of team members in table
- Columns: Name, Email, Role, Joined
- "Invite Member" button (shows modal with email input)
- Remove member button

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## BILLING PAGE (BillingPage.tsx)

3 pricing plan cards:

Starter — £49/month
- Up to 3 social accounts
- AI content generation
- Content calendar
- Manual posting
- Basic analytics
- Email support

Growth — £99/month ⭐ POPULAR
- Everything in Starter
```

```
- Instagram + Facebook integration
- Advanced analytics
- Hashtag suggestions
- 5 team members
- Priority support

Pro — £149/month
- Everything in Growth
- TikTok integration
- Auto-scheduling
- AI hashtag generator
- Unlimited team members
- White-label reports

Enterprise — £299/month
- Everything in Pro
- AI Agent (daily posts, auto-scheduling, comment replies)
- Monthly AI reports
- WhatsApp/email notifications
- Dedicated account manager

Current plan highlighted in blue. "Upgrade" buttons on other plans.
"Powered by Stripe" note at bottom (Stripe integration as future phase).

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## UI DESIGN SYSTEM

Brand colors:
- Primary: #2563EB (blue)
- Primary dark: #1D4ED8
- Secondary: #7C3AED (purple) — used for AI features
- Success: #10B981 (green)
- Warning: #F59E0B (amber)
- Danger: #EF4444 (red)
- Background: #F8FAFC (light grey)
- Sidebar: #0F172A (near black)
- Card: #FFFFFF
- Border: #E2E8F0
- Text primary: #1E293B
- Text muted: #64748B

Typography:
- Font: Inter (Google Fonts) — clean, modern, SaaS-appropriate
- Headings: font-semibold or font-bold
- Body: font-normal, text-sm or text-base
- Stats/numbers: font-bold, tabular nums
```

Border radius: rounded-xl for cards, rounded-lg for inputs, rounded-full for badges
Shadows: shadow-sm for cards, shadow-md on hover

All pages must be fully mobile responsive:
- Sidebar collapses to bottom nav bar on mobile
- Stats stack vertically
- Tables become card lists on mobile
- Use Tailwind responsive prefixes: sm: md: lg: xl:

DEMO SEED DATA (supabase/seed.sql)

Insert a demo company and sample data so the app looks populated on first login:

```sql
-- Demo company
INSERT INTO companies (id, company_name, industry, location, plan) VALUES
('demo-company-uuid', 'Senlac Guest Houses', 'Hotel / Guest House', 'St
Leonards-on-Sea, East Sussex', 'growth');

-- Demo social accounts
INSERT INTO social_accounts (company_id, platform, username, followers,
posts_count, engagement_rate, connected) VALUES
('demo-company-uuid', 'instagram', '@senlacguesthouses', 1240, 35, 5.8,
true),
('demo-company-uuid', 'facebook', 'Senlac Guest Houses', 2130, 48, 6.1,
true),
('demo-company-uuid', 'tiktok', '@senlacguesthouses', 890, 21, 4.7, true);

-- Demo analytics (last 7 days for each platform)
-- Insert 7 rows per platform with realistic growing numbers

-- Demo posts (mix of published, scheduled, draft)
-- Insert 8 sample posts with captions and engagement data
```

ADDITIONAL REQUIREMENTS

1. Loading states: Every data fetch must show a skeleton loader, not a blank screen.

2. Error handling: All API calls wrapped in try/catch. Show toast notifications for success/error using a toast component.

3. Empty states: When no data exists (no posts, no media), show friendly empty state with icon, message, and action button.

4. Toast notifications: Use a toast/notification system for: post saved, post scheduled, content copied, settings saved, error messages.

5. Responsive sidebar: On desktop: full sidebar visible. On tablet: icon-only sidebar. On mobile: hidden sidebar with hamburger menu button that shows overlay nav.

6. Theme: Light mode only for Phase 1. Dark mode as future feature.

7. All charts use Recharts with animated transitions.

8. All forms use controlled components with proper TypeScript types.

9. All Supabase queries include company_id filter as backup to RLS.

10. API calls from frontend go to Express server (not directly to OpenAI — key stays server-side).

11. Create a comprehensive README.md with:
    - Project overview
    - Local development setup
    - Supabase setup steps (with exact SQL to run)
    - Environment variable reference
    - Deployment guide for Vercel
    - Architecture diagram in ASCII art

12. .gitignore must include: node_modules, .env, dist, .DS_Store

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PACKAGE.JSON SCRIPTS

Root package.json (monorepo):
```json
{
  "scripts": {
    "dev": "concurrently \"npm run dev:client\" \"npm run dev:server\"",
    "dev:client": "cd client && npm run dev",
    "dev:server": "cd server && npm run dev",
    "build": "cd client && npm run build",
    "install:all": "npm install && cd client && npm install && cd ../server
```

```
&& npm install"
  }
}
```

FINAL OUTPUT CHECKLIST

Ensure ALL of the following are created and functional:

✅ Supabase schema SQL file
✅ RLS policies SQL file
✅ Seed data SQL file
✅ Auth trigger SQL
✅ Express server with all 4 route files
✅ OpenAI service with proper prompts
✅ React app with all 10 pages
✅ AuthContext with session persistence
✅ Protected routes
✅ Sidebar navigation
✅ Dashboard with stats + charts + social cards + AI panel
✅ AI Generator with all 4 output types + calendar generator
✅ Content Studio with preview
✅ Analytics page with 4 chart types
✅ Calendar page (month + week view)
✅ Media Library with upload
✅ Social Accounts page
✅ Settings page with all tabs
✅ Billing page with 4 plans
✅ .env.example
✅ README.md with setup instructions
✅ Mobile responsive design
✅ Loading skeletons
✅ Toast notifications
✅ Error handling
✅ Empty states
✅ TypeScript types for all data models

Build this application now. Start with the folder structure, then the database files, then the server, then the React client. Make it production-ready, clean, and maintainable.

AFTER REPLIT BUILDS — SETUP CHECKLIST

Step 1: Supabase Setup

1. Go to supabase.com → New Project
2. Run `supabase/schema.sql` in the SQL editor
3. Run `supabase/rls.sql` in the SQL editor
4. Run `supabase/seed.sql` in the SQL editor (optional demo data)
5. Go to Storage → create bucket named `media` → set to Public
6. Copy your Project URL and anon key

Step 2: Environment Variables

Set these in Replit Secrets:

```
VITE_SUPABASE_URL = https://xxxx.supabase.co
VITE_SUPABASE_ANON_KEY = eyJ...
VITE_API_BASE_URL = https://your-replit-server-url
SUPABASE_URL = https://xxxx.supabase.co
SUPABASE_SERVICE_ROLE_KEY = eyJ...
OPENAI_API_KEY = sk-...
PORT = 3001
```

Step 3: First Login

1. Sign up with your email
2. Check Supabase → Auth → Users to confirm account created
3. Check Supabase → profiles table to confirm trigger worked
4. Log in and explore the dashboard

Step 4: Deploy to Vercel

1. Push code to GitHub
2. Import repo in Vercel
3. Set root directory to `client`
4. Add all VITE_ environment variables
5. Deploy

PHASE ROADMAP

Phase	Features	Price	Timeline
✅ Phase 1	Auth, Dashboard, AI Generator, Content Studio, Calendar, Media	£49/mo	Now
🔜 Phase 2	Instagram API, Facebook API, Real Analytics	£99/mo	Month 2
🔜 Phase 3	TikTok API, Auto-scheduling, AI Hashtags, Review Management	£149/mo	Month 3
🔜 Phase 4	AI Agent, Daily posts, WhatsApp reports, Comment replies	£299/mo	Month 5

Built for LE Software Solutions — LE Growth AI is the second product in the LE SaaS suite following LeHR.